Matris ve Özellikleri

Matris: $p \times r$ tane sayının p satır ve r sütuna yerleştirilmesiyle oluşan tabloya matris denir.

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1r} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2r} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{p1} & a_{p2} & a_{p3} & \cdots & a_{pr} \end{bmatrix}_{p \times r}$$

p satır

r sütun

A matrisi aynca $1 \le i \le p$, $1 \le j \le r$ olmak üzere $A = [a_{ij}]_{p \times r}$ olarak yazılır.

$(A_{ij} \in R)$

* **Eşit Matrisler:** Aynı satır ve sütuna sahip $A = [a_{ij}]_{p \times r}$ ve $B = [b_{ij}]_{p \times r}$ matrisleri için $\forall\, 1 \le i \le p$, $1 \le j \le r$ için

$$a_{ij} = b_{ij} \rightarrow A = B \text{ dir.}$$

* $A = \begin{bmatrix} 1 & 0 & -1 \\ x+y & 2 & 0 \\ -2 & 1 & z \end{bmatrix}$ ve $B = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 2 & 0 \\ x-y & 1 & 1 \end{bmatrix}$, $A = B$ ise $x, y, z = ?$

$$\begin{array}{r} x+y = 0 \\ x-y = -2 \\ \hline 2x = -2 \\ x = -1 \end{array} \qquad \begin{array}{l} z = 1 \\ x = -1 \\ y = 1 \end{array}$$

- $p=r=1 \Rightarrow A=[a_{11}]_{1\times 1}$
- $p=1 \Rightarrow A=[\,\cdots\,]_{1\times r}$
- $r=1 \Rightarrow A=\begin{bmatrix}\vdots\\ \vdots\\ \vdots\end{bmatrix}_{p\times 1}$

* **Kare Matris:** $p=r$ old. $A_{p\times r}$ matrisine kare matris denir.

* **Köşegen Matris:** $A_{p\times p}$ kare matrisi için a_{ii}, $1\le i\le p$ elemanlarına köşegen elemanlar denir. Köşegen elemanları dışında kalan elemanları sıfır olan matrise köşegen matris denir.

$$A=\begin{bmatrix} a_{11} & 0 & 0 & 0\\ 0 & a_{22} & 0 & 0\\ 0 & 0 & a_{33} & 0\\ 0 & 0 & 0 & \ddots \end{bmatrix}$$

* **Sıfır Matris:** Tüm elemanları sıfır (0) olan matrise denir.

$$\begin{bmatrix}0&0\\0&0\end{bmatrix},\ [0],\ \begin{bmatrix}0\\0\\0\end{bmatrix},\ \cdots$$

$$A.A+2.A=O_{3\times 3}$$

$x^2+2x=0$

$x=M_{3\times 3}$ için $[\,]_{2\times 2}[\,]=0$

∝ **Birim Matris:** Bir köşegen matrisin tüm köşegen elemanları I ise denir.

$$I_{2x2} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$I_{pxp} = (\delta_{ij})_{pxp} = \delta_{ij} = \begin{cases} 1, & i=j \\ 0, & i \neq j \end{cases}$$

(Kronecker delta)

∝ **Skaler Matris:** Bir köşegen matrisin sıfırdan farklı tüm elemanları eşit ise denir.

$$k \cdot I_{2x2} = \begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix}, \quad 3 \cdot I_{3x3} = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

Matris Toplamı

$A = [a_{ij}]_{mxn}$ ve $B = [b_{ij}]_{mxn}$ aynı sıralı iki matris olsun. $A + B = C \Rightarrow C = [c_{ij}]_{mxn}$ $c_{ij} = a_{ij} + b_{ij}$ olacak şekilde C matrisine A ve B'nin toplamı denir.

$$A + B = C$$

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{m1} & b_{m2} & \cdots & b_{mn} \end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{m1} & c_{m2} & \cdots & c_{mn} \end{bmatrix}$$

Matris Toplamının Özellikleri:

1) Değişme Özelliği

$\forall A, B \in M_{m\times n}$ için $A + B = B + A$

2) Birleşme Özelliği

$\forall A, B, C \in M_{m\times n}$ için $(A + B) + C = A + (B + C)$

3) Etkisiz Eleman

$\forall A, B, C \in M_{m\times n}$ için $A + O = O + A = A$

O matrisi, matris toplam işleminin etkisiz elemanıdır.

Matrislerde Skaler Çarpma

$A \in M_{m\times n}$ ve $r \in R$ için $b_{ij} = r.a_{ij}$ olan B matrisi A matrisinin r skaleriyle çarpılmış halidir. $B = r.A$

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}_{m\times n} \qquad r.A = \begin{bmatrix} ra_{11} & ra_{12} & \cdots & ra_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ ra_{m1} & ra_{m2} & \cdots & ra_{mn} \end{bmatrix}_{m\times n}$$

Skaler Çarpma İşleminin Özellikleri:

$\forall A, B \in M_{m\times n}$ ve $r, s \in R$ için

1) $(r + s)A = rA + sA$

2) $r(A+B) = rA + rB$

3) $(rs)A = r(sA)$ (sağ: skaler, skaler)

4) $1.A = A$

5) $0.A = 0_{m\times n}$

· A matrisinin toplamsal tersi $-A$'dır.

İki Matrisin Çarpımı: Vize

$A = [a_{ij}]_{m\times p}$ ve $B = [b_{ij}]_{p\times n}$ için $C = A.B$ matrisi

$c_{ij} = \sum_{k=1}^{p} a_{ik} b_{kj}$ olarak tanımlanır.

· $A_{m\times p} \cdot B_{p\times n} = C_{m\times n}$

$$A \times B = C$$

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1p} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mp} \end{bmatrix} \times \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{p1} & b_{p2} & \cdots & b_{pn} \end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{m1} & c_{m2} & \cdots & c_{mn} \end{bmatrix}$$

$$c_{11} = \sum_{k=1}^{p} a_{1k} b_{k1} = a_{11}b_{11} + a_{12}b_{21} + \cdots + a_{1p}b_{p1}$$

a) $A = \begin{bmatrix} -1 & 0 \\ 1 & 3 \end{bmatrix}_{2\times 2}$

$B = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{bmatrix}_{2\times 7}$

$A_{2\times 2} \times B_{2\times 7} = C_{2\times 7}$

$A \times B = \begin{bmatrix} & & & & & & \\ & & & & & & \end{bmatrix}$

c_{24}

$1.4 + 3.4 = 16$

Matris Çarpımının Özellikleri:

1) Birleşme özelliği vardır.

$A_{m\times p}, B_{p\times q}, C_{q\times n}$ için $(A.B)C = A.(B.C)$

2) $A_{m\times p}$, $B_{p\times q}$, $C_{p\times q}$ olsun. $A(B+C) = AB + AC$

3) $A_{m\times p}$ ve $B_{p\times q}$ için $(rA)(sB) = (rs)(AB)$

4) $AI = IA = A$

5) Değişme özelliği yoktur. $AB \neq BA$ (Her zaman eşit değil)

∝ $A = \begin{bmatrix} -1 & 2 \\ 1 & 0 \end{bmatrix}$ ve $B = \begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix}$ olsun. $AB \stackrel{?}{=} BA$

$$AB = \begin{bmatrix} -1 & 2 \\ 1 & 0 \end{bmatrix} \times \begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} c_{11} = -1.3+2.2 = 1 & c_{12} = -1.1+2.4 = 7 \\ c_{21} = 3 & c_{22} = 1 \end{bmatrix}$$

$$BA = \begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix} \times \begin{bmatrix} -1 & 2 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} c_{11} = -2 & c_{12} = 6 \\ c_{21} = 2 & c_{22} = 4 \end{bmatrix}$$

$$AB \neq BA$$

∝ $\underset{2\times 3}{\begin{bmatrix} 1 & 2 & -1 \\ 3 & 1 & 4 \end{bmatrix}} \times \underset{3\times 2}{\begin{bmatrix} -2 & 5 \\ 4 & -3 \\ 2 & 1 \end{bmatrix}} = ? \Rightarrow \underset{2\times 2}{\begin{bmatrix} c_{11} = -2+8-2 = 4 & c_{12} = 5-6-1 = -2 \\ c_{21} = 18 & c_{21} = 16 \end{bmatrix}}$

$$\alpha \underset{1\times3}{\begin{bmatrix} 7 & -4 & 5 \end{bmatrix}} \times \underset{3\times1}{\begin{bmatrix} 3 \\ 2 \\ -1 \end{bmatrix}} = \underset{1\times1}{\begin{bmatrix} c_{11} = 8 \end{bmatrix}}$$

$$\alpha \underset{3\times1}{\begin{bmatrix} 3 \\ 2 \\ -1 \end{bmatrix}} \times \underset{1\times3}{\begin{bmatrix} 7 & -4 & 5 \end{bmatrix}} = \underset{3\times3}{\begin{bmatrix} c_{11}= & c_{12}= & c_{13}= \\ c_{21}= & c_{22}=-8 & c_{23}= \\ c_{31}= & c_{32}=4 & c_{33}= \end{bmatrix}}$$

$$\alpha \; A = \underset{2\times2}{\begin{bmatrix} 2 & 0 \\ 1 & -1 \end{bmatrix}}$$

$$B = \underset{3\times2}{\begin{bmatrix} -1 & 1 \\ 0 & 1 \\ 2 & 3 \end{bmatrix}} \quad \text{Tính} \quad B \cdot A = ?$$

$$\underset{3\times2}{\begin{bmatrix} -1 & 1 \\ 0 & 1 \\ 2 & 3 \end{bmatrix}} \times \underset{2\times2}{\begin{bmatrix} 2 & 0 \\ 1 & -1 \end{bmatrix}} = \underset{3\times2}{\begin{bmatrix} c_{11}=-1 & c_{12}=-1 \\ c_{21}= & c_{22}= \\ c_{31}=7 & c_{32}= \end{bmatrix}}$$

$$\alpha \; A = \underset{2\times3}{\begin{bmatrix} 1 & x & 3 \\ 2 & -1 & 1 \end{bmatrix}}$$

$$B = \underset{3\times1}{\begin{bmatrix} 2 \\ 4 \\ y \end{bmatrix}} \quad \text{Tìm} \quad AB = \begin{bmatrix} 12 \\ 6 \end{bmatrix} \Rightarrow x+y = ?$$

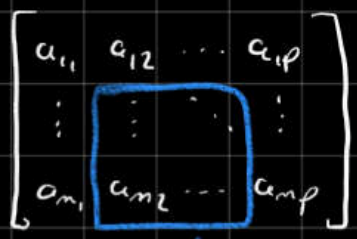
$$AB = \underset{2\times3}{\begin{bmatrix} 1 & x & 3 \\ 2 & -1 & 1 \end{bmatrix}} \times \underset{3\times1}{\begin{bmatrix} 2 \\ 4 \\ y \end{bmatrix}} = \underset{2\times1}{\begin{bmatrix} c_{11}=2+4x+3y \\ c_{21}=4-4+y \end{bmatrix}} = \begin{bmatrix} 12 \\ 6 \end{bmatrix}$$

$y=6, x=-2 \quad x+y=4$

Alt Matris

Bir $A_{m \times n}$ matrisinin elemanlarını yerlerini koruyarak içeren A'dan küçük matrise A'nın alt matrisi denir.

A



↳ A'nın alt m.

A'nın Blok Matris olarak Gösterilmesi

A matrisi uygun boyutlu alt matrislerle bloklara ayrılıp gösterilebilir. Bu gösterime blok matris gösterimi denir.

$$A = \left[\begin{array}{cc|c} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ \hline a_{31} & a_{32} & a_{33} \end{array}\right] = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$$

$$A_{11} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \qquad A_{12} = \begin{bmatrix} a_{13} \\ a_{23} \end{bmatrix}$$

$$A_{21} = \begin{bmatrix} a_{31} & a_{32} \end{bmatrix} \qquad A_{22} = \begin{bmatrix} a_{33} \end{bmatrix}$$

Blok Matrisler ile Çarpma

$A_{m \times p}$ ve $B_{p \times q}$ matrisleri uygun boyutlu blok matrislere ayrılarak blok matris çarpımı yapılabilir.

α $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}_{2\times 2}$ $\quad B = \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}_{2\times 2}$

$$A.B = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}_{2\times 2} \times \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}_{2\times 2}$$

$$= \begin{bmatrix} A_{11}B_{11} + A_{12}B_{21} = 4 & A_{11}B_{12} + A_{12}B_{22} = \\ A_{21}B_{11} + A_{22}B_{21} = & A_{21}B_{12} + A_{22}B_{22} = \end{bmatrix}_{2\times 2}$$

$$\begin{bmatrix} 1 & 2 \end{bmatrix} \times \begin{bmatrix} -2 \\ 4 \end{bmatrix} + [-1][2]$$

$$-2 + 8 = 6 \quad + \quad -2 \quad = 4$$

α $A_{p\times q}$, $B_{q\times r}$, $C_{r\times p}$ boyutlu matrisler için,

a) $AB = p\times q \cdot q\times r \rightarrow \checkmark \rightarrow p\times r$

b) $BC = q\times r \cdot r\times p \rightarrow \checkmark \rightarrow q\times p$

c) $BAC = q\times r \cdot p\times q \rightarrow X$

d) $ABC = p\times q \cdot q\times r \cdot r\times p \rightarrow \checkmark \rightarrow p\times p$

e) $BCA = \rightarrow \checkmark \rightarrow q\times q$

α $A = \begin{bmatrix} 1 & 2 \\ -1 & 0 \end{bmatrix}_{2\times 2}$ iken $A^2 - 2A + 1$ işlemini hesaplayın.

$x^2 - 2x + 1$ denkleminde $x = A$ matrisi

$$\begin{bmatrix} 1 & 2 \\ -1 & 0 \end{bmatrix}_{2\times 2} \cdot \begin{bmatrix} 1 & 2 \\ -1 & 0 \end{bmatrix}_{2\times 2} - 2 \cdot \begin{bmatrix} 1 & 2 \\ -1 & 0 \end{bmatrix}_{2\times 2} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}_{2\times 2}$$

$$\begin{bmatrix}-1 & 2\\ -1 & -2\end{bmatrix}_{2\times2} + \begin{bmatrix}-2 & -4\\ 2 & 0\end{bmatrix} + \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix} = \begin{bmatrix}-1-2+1 & 2-4+0\\ -1+2+0 & -2+0+1\end{bmatrix} = \begin{bmatrix}-2 & -2\\ 1 & -1\end{bmatrix}_{2\times2}$$

$A^2 \quad -2A \quad + \quad I$

α $A=[a_{ij}]_{3\times3}$ matrisleri yandaki şekilde tanımlanıyor.

$B=[b_{ij}]_{3\times3}$ $\quad a_{ij}=(i-j) \quad b_{ij}=(i.j)$ buna göre $A.B=?$

$$A=\begin{bmatrix}a_{11} & a_{12} & a_{13}\\ a_{21} & a_{22} & a_{23}\\ a_{31} & a_{32} & a_{33}\end{bmatrix}_{3\times3}$$

$a_{11}=(1-1)=0 \qquad a_{21}=2-1=1$

$a_{12}=(1-2)=-1$

$a_{13}=(1-3)=-2$

$$B=\begin{bmatrix}b_{11} & b_{12} & b_{13}\\ b_{21} & b_{22} & b_{23}\\ b_{31} & b_{32} & b_{33}\end{bmatrix}_{3\times3}$$

$b_{11}=(1.1)=1 \qquad b_{21}=2.1=2$

$b_{12}=(1.2)=2$

$b_{13}=(1.3)=3$

$$\begin{bmatrix}0 & -1 & -2\\ 1 & 0 & -1\\ 2 & 1 & 0\end{bmatrix}_{3\times3} \cdot \begin{bmatrix}1 & 2 & 3\\ 2 & 4 & 6\\ 3 & 6 & 9\end{bmatrix}_{3\times3} = \begin{bmatrix} & c_{12} & \\ & & c_{23}\\ c_{31} & & \end{bmatrix}_{3\times3}$$

$A \quad . \quad B \quad = \quad C$

$\rightarrow c_{12}=0.2-1.4-2.6=-16$

$c_{23}=3+0-9=-6$

$c_{31}=2+2+0=4$

Bir Matrisin Transpozu

$A_{m\times n}$ lik bir matris ve $A=[a_{ij}]_{m\times n}$ olsun. A'nın transpozu A^T veya A^t ile gösterilir.

$$A^t=[A_{ji}]_{n\times m}$$

$$A = [\,a_{11}\ a_{12}\ a_{13}\,]_{1\times 3} \Rightarrow A^t = \begin{bmatrix} a_{11} \\ a_{21} \\ a_{31} \end{bmatrix}_{3\times 1}$$

Özellikler: $A_{m\times n}$ lik bir matris olsun ve $r \in R$ için

- $(A^t)^t = A$
- $(A+B)^t = A^t + B^t$
- $(rA)^t = r(A)^t$
- $(AB)^t = B^t \cdot A^t$

Simetrik Matris: $A_{n\times n}$ lik matrise simetrik matris denir.

$\rightarrow A^t = A$

- Simetrik matrisler kare formdadır.

Anti Simetrik Matris: $A_{n\times n}$ lik matris antisimetrik matristir.

$\rightarrow A^t = -A$

Bir Matrisin İzi: $A_{n\times n}$ lik kare bir matris olsun. A matrisinin izi $iz(A)$ veya $tr(A)$ (trace) olarak gösterilir.

$\rightarrow iz(A) = a_{11} + a_{22} + \cdots a_{nn}$

Özellikler:

- $iz(A) + iz(B) = iz(A+B)$
- $iz(cA) = c(iz(A))$
- $iz(A) = iz(A^t)$
- $iz(AB) = iz(BA)$

"A matrisi idempotenttir" $\iff$ "$A^2 = A$"

$A = \begin{bmatrix} 1 & x \\ 2 & x \end{bmatrix}$ ve A idempotent ise $x = ?$

$$\begin{bmatrix} 1 & x \\ 2 & x \end{bmatrix} \cdot \begin{bmatrix} 1 & x \\ 2 & x \end{bmatrix} = \begin{bmatrix} 1 & x \\ 2 & x \end{bmatrix}$$

$$\begin{bmatrix} 1+2x & x+x^2 \\ 2+2x & 2x+x^2 \end{bmatrix} = \begin{bmatrix} 1 & x \\ 2 & x \end{bmatrix}$$

$x = 0$

"A matrisi nilpotenttir" $\iff$ "$A^n = 0$", $n \in \mathbb{Z}^+$

$A = \begin{bmatrix} 0 & x & y \\ 0 & 0 & z \\ 0 & 0 & 0 \end{bmatrix}_{3\times 3}$ matrisinin nilpotent old. gösterin.

$\underbrace{A \cdot A \cdots A}_{n \text{ tane}} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$$\begin{bmatrix} 0 & x & y \\ 0 & 0 & z \\ 0 & 0 & 0 \end{bmatrix}_{3\times 3} \cdot \begin{bmatrix} 0 & x & y \\ 0 & 0 & z \\ 0 & 0 & 0 \end{bmatrix}_{3\times 3} = \begin{bmatrix} 0 & 0 & xz \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}_{3\times 3} = A^2$$

$$\begin{bmatrix} 0 & 0 & xz \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}_{3\times 3} \cdot \begin{bmatrix} 0 & x & y \\ 0 & 0 & z \\ 0 & 0 & 0 \end{bmatrix}_{3\times 3} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}_{3\times 3} = A^3 = 0$$

Alt Üçgensel Matris: $A_{n\times n}$ lik kare bir matris olsun. Eğer A matrisinin köşegen üstündeki tüm elemanları sıfır ise A'ya alt üçgensel m. denir.

$$A = \begin{bmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Üst Üçgensel Matris: $A_{n\times n}$ lik kare bir matris olsun. Eğer A matrisinin köşegen altındaki tüm elemanları sıfır ise A'ya üst üçgensel m. denir.

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{bmatrix}$$

Vize

Temel Satır İşlemleri: Bir A matrisinin satır (veya sütun) üzerinde yapılan aşağıdaki işlemlere temel işlemler denir. (Satır = Sütun)

- Bir matrisin iki satırını kendi aralarında değiştirmek.
  $r_i \leftrightarrow r_j$ (i. satır ile j. satır değişimi)
- Bir matrisin i. satırını $k \in R$ ile çarpmak. $(kr_i \rightarrow r_i)$
- Bir matrisin i. satırını $k \in R^+$ ile çarpıp j. satıra eklemek.
  $kr_i + r_j \rightarrow r_j$

Denk Matrisler: Temel satır işlemleri ile birbirine dönüştürülebilen matrislerdir.

∝ $A = \begin{bmatrix} 2 & -1 & 3 \\ 0 & -1 & 0 \\ 2 & 1 & 6 \end{bmatrix}$ matrisinin I'ya denk old. göster.

$$\begin{bmatrix} 2 & -1 & 3 \\ 0 & -1 & 0 \\ 2 & 1 & 6 \end{bmatrix} \xrightarrow[-1.r_2+r_1 \to r_1]{r_2+r_3 \to r_3} \begin{bmatrix} 2 & 0 & 3 \\ 0 & -1 & 0 \\ 2 & 0 & 6 \end{bmatrix} \xrightarrow[-r_1+r_3 \to r_3]{-r_2 \to r_2} \begin{bmatrix} 2 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$\downarrow \tfrac{1}{3} r_3 \to r_3$$

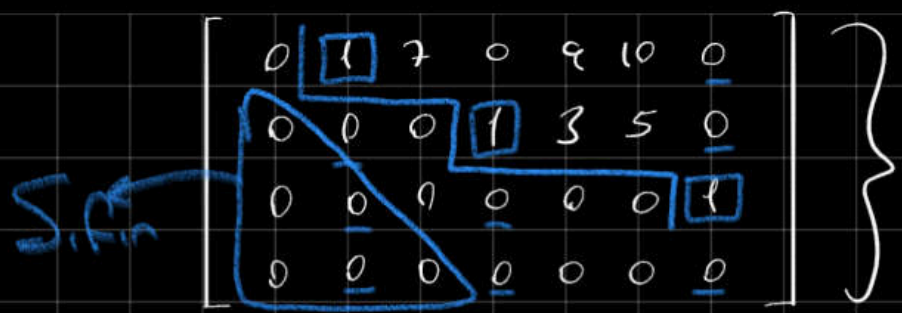
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xleftarrow{\frac{1}{2} r_1 \to r_1} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xleftarrow{-3r_3+r_1 \to r_1} \begin{bmatrix} 2 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Vize

Satırca İndirgenmiş Basamak Formu: $A_{m \times n}$ lik matris aşağıdaki özelliklere sahip bir matrise elementer işlemler yardımıyla dönüştürülebiliyorsa A matrisine denk gelen satırca indirgenmiş basamak matris elde edilmiş olur.

1) Sıfır satırı varsa en altta olmalı
2) Soldan sıfırdan farklı ilk eleman 1 olmalı (Pivot)
3) Sıfır satırı hariç diğer satırların pivotları basamak halinde olmalı.
4) Pivotun bulunduğu sütundaki tüm elemanlar sıfır olmalı.

· 4 hariç ilk 3 şart sağlanıyorsa bu matrise basamak form denir. (indirgenmiş)



ö $A = \begin{bmatrix} 1 & 3 & 5 & -2 \\ 1 & 4 & 6 & -2 \\ -1 & -1 & -3 & 2 \end{bmatrix}_{3\times4}$;

i) A matrisini basamak hale getirin.

Pivot

$$\begin{bmatrix} 1 & 3 & 5 & -2 \\ 1 & 4 & 6 & -2 \\ -1 & -1 & -3 & 2 \end{bmatrix}_{3\times4} \xrightarrow[-r_1+r_3\to r_3]{-r_1+r_2\to r_2} \begin{bmatrix} 1 & 3 & 5 & -2 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & 2 & 0 \end{bmatrix}_{3\times4} \xrightarrow{-2r_2+r_3\to r_3} \begin{bmatrix} 1 & 3 & 5 & -2 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}_{3\times4}$$

ii) A matrisini indirgenmiş basamak hale getirin.

$$\begin{bmatrix} 1 & 3 & 5 & -2 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}_{3\times4} \xrightarrow{-3r_2+r_1\to r_1} \begin{bmatrix} 1 & 0 & 2 & -2 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}_{3\times4}$$

ö $A = \begin{bmatrix} 0 & 3 & 4 \\ 2 & -1 & 6 \\ -1 & 2 & 5 \end{bmatrix}_{3\times3}$ A'nın satırca indirgenmiş bas. f. bulunuz.

$$\begin{bmatrix} 0 & 3 & 4 \\ 2 & -1 & 6 \\ -1 & 2 & 5 \end{bmatrix}_{3\times3} \xrightarrow{r_1\leftrightarrow r_3} \begin{bmatrix} -1 & 2 & 5 \\ 2 & -1 & 6 \\ 0 & 3 & 4 \end{bmatrix}_{3\times3} \xrightarrow{2r_1+r_2\to r_2} \begin{bmatrix} -1 & 2 & 5 \\ 0 & 3 & 16 \\ 0 & 3 & 4 \end{bmatrix}_{3\times3} \xrightarrow{-r_2+r_3\to r_3} \begin{bmatrix} 1 & -2 & -5 \\ 0 & 3 & 16 \\ 0 & 0 & -12 \end{bmatrix}_{3\times3}$$

$\xrightarrow{\frac{-1}{11}r_3 \to r_3} \begin{bmatrix} 1 & -2 & -5 \\ 0 & 3 & 16 \\ 0 & 0 & 1 \end{bmatrix}_{3\times3} \xrightarrow[5r_3+r_1 \to r_1]{-16r_3+r_2 \to r_2} \begin{bmatrix} 1 & -2 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{3\times3} \xrightarrow{\frac{1}{3}r_2 \to r_2} \begin{bmatrix} 1 & -2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{3\times3}$

$\xrightarrow{2r_2+r_1 \to r_1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{3\times3}$

α

$A = \begin{bmatrix} 1 & 1 & 1 & 9 \\ 0 & 1 & -2 & 5 \\ 0 & 1 & 2 & 2 \\ 0 & 2 & 0 & 7 \end{bmatrix}_{4\times4} \xrightarrow[-2r_2+r_4 \to r_4]{-r_2+r_3 \to r_3} \begin{bmatrix} 1 & 1 & 1 & 9 \\ 0 & 1 & -2 & 5 \\ 0 & 0 & 4 & -3 \\ 0 & 0 & 4 & -3 \end{bmatrix} \xrightarrow{-r_3+r_4 \to r_4} \begin{bmatrix} 1 & 1 & 1 & 9 \\ 0 & 1 & -2 & 5 \\ 0 & 0 & 4 & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

$\xrightarrow[\frac{1}{4}r_3 \to r_3]{-r_2+r_1 \to r_1} \begin{bmatrix} 1 & 0 & 3 & 4 \\ 0 & 1 & -2 & 5 \\ 0 & 0 & 1 & -3/4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow[-3r_3+r_1 \to r_1]{2r_3+r_2 \to r_2} \begin{bmatrix} 1 & 0 & 0 & 25/4 \\ 0 & 1 & 0 & 7/2 \\ 0 & 0 & 1 & -5/4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

Bir Matrisin Tersi

A, nxn lik kare matris olsun $AB = I_n$ ve $BA = I_n$ şartını sağlayan B matrisine A matrisinin ters denir. (çarpımsal ters) $A \cdot A' = I$

α $A = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}_{2\times2}$ $B = \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix}_{2\times2}$

$AB = \begin{bmatrix} \underset{1}{3.1+2.-1} & \underset{0}{3.-1+2.3} \\ \underset{0}{1.1+1.-1} & \underset{1}{1.-2+1.3} \end{bmatrix}_{2\times2}$ $BA = \begin{bmatrix} 1.3+(-2).-1 & 1.2+1.-2 \\ -1.3+3.1 & -1.2+3.1 \end{bmatrix}_{2\times2}$

α $A=\begin{bmatrix}1 & 0\\ -1 & 1\end{bmatrix}_{2x2}$ matrisin tersini bulunuz.

$$\begin{bmatrix}1 & 0\\ -1 & 1\end{bmatrix}\cdot\begin{bmatrix}x & y\\ z & t\end{bmatrix}=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}=\begin{bmatrix}1.x & 1.y\\ -x+z & -y+t\end{bmatrix}_{2x2}$$

$$A \cdot A' = I$$

$x=1\ \ y=0$
$z=1\ \ t=1$ $\Big\}\ A^{-1}\begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}_{2x2}$

Temel İşlemlerle Matrisin Tersi Vize

Tersi olan A matrisi için $[A\,|\,I]$ eki matris oluşturulur. A matrisi temel işlemlerle birim matrise dönüştürüldüğünde elde edilen ek matris A^{-1} dir.

$$[A\,|\,I]\rightarrow[I\,|\,A^{-1}]$$

α $A=\begin{bmatrix}2 & 1 & 1\\ 0 & 1 & 1\\ 2 & 0 & 1\end{bmatrix}_{3x3}$ için $A^{-1}=?$

$$\left[\begin{array}{ccc|ccc}2 & 1 & 1 & 1 & 0 & 0\\ 0 & 1 & 1 & 0 & 1 & 0\\ 2 & 0 & 1 & 0 & 0 & 1\end{array}\right]\rightarrow[I\,|\,A^{-1}]$$ 1.

$$\left[\begin{array}{ccc|ccc}2 & 1 & 1 & 1 & 0 & 0\\ 0 & 1 & 1 & 0 & 1 & 0\\ 2 & 0 & 1 & 0 & 0 & 1\end{array}\right]\xrightarrow{r_1+r_3\rightarrow r_3}\left[\begin{array}{ccc|ccc}-2 & 1 & 1 & 1 & 0 & 0\\ 0 & 1 & 1 & 0 & 1 & 0\\ 0 & 1 & 2 & 1 & 0 & 1\end{array}\right]$$ 2.

$$\xrightarrow[-r_2+r_1\to r_1]{-r_2+r_3\to r_3} \left[\begin{array}{ccc|ccc} -2 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{array}\right] \xrightarrow[-r_3+r_2\to r_2]{\frac{-1}{2}r_1\to r_1} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 1 & 0 & -1 & 2 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{array}\right]$$

A^{-1} 13.

a) $A = \left[\begin{array}{ccc} 2 & 3 & 4 \\ 0 & 1 & 1 \\ -1 & 1 & 1 \end{array}\right]$ TSE $A^{-1} = ?$

$$\left[\begin{array}{ccc} 2 & 3 & 4 \\ 0 & 1 & 1 \\ -1 & 1 & 1 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|ccc} 2 & 3 & 4 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ -1 & 1 & 1 & 0 & 0 & 1 \end{array}\right] \xrightarrow{r_1 \leftrightarrow r_3} \left[\begin{array}{ccc|ccc} -1 & 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ -2 & 3 & 4 & 1 & 0 & 0 \end{array}\right]$$

$$\xrightarrow[-3r_2+r_3\to r_3]{-r_2+r_1\to r_1} \left[\begin{array}{ccc|ccc} -1 & 0 & 0 & 0 & -1 & 1 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 2 & 0 & 1 & 1 & -3 & 0 \end{array}\right] \xrightarrow{-r_1\to r_1} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 1 & -1 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 2 & 0 & 1 & 1 & -3 & 0 \end{array}\right]$$

$$\xrightarrow{-2r_1+r_3\to r_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 1 & -1 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & -5 & 2 \end{array}\right] \xrightarrow{-r_3+r_2\to r_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 1 & -1 \\ 0 & 1 & 0 & -1 & 6 & -2 \\ 0 & 0 & 1 & 1 & -5 & 2 \end{array}\right]$$

I A^{-1}

b) $A = \left[\begin{array}{cccc} 1 & -2 & 3 & 4 \\ 2 & 1 & 1 & 3 \\ 3 & -1 & 2 & 1 \end{array}\right]_{3\times 4}$ A^{-1}'yi satırca indirgeyiniz.

$$\begin{bmatrix} 1 & -2 & 3 & 4 \\ 2 & 1 & 1 & 3 \\ 3 & -1 & 2 & 1 \end{bmatrix} \xrightarrow{-2r_1+r_2\to r_2} \begin{bmatrix} 1 & -2 & 3 & 4 \\ 0 & 5 & -5 & -5 \\ 3 & -1 & 2 & 1 \end{bmatrix} \xrightarrow{-3r_1+r_3\to r_3} \begin{bmatrix} 1 & -2 & 3 & 4 \\ 0 & 5 & -5 & -5 \\ 0 & 5 & -7 & -11 \end{bmatrix}$$

1. adım, 2. adım

$$\xrightarrow{-r_2+r_3\to r_3} \begin{bmatrix} 1 & -2 & 3 & 4 \\ 0 & 5 & -5 & -5 \\ 0 & 0 & -2 & -6 \end{bmatrix} \xrightarrow[-\frac{1}{2}r_3\to r_3]{\frac{1}{5}r_2\to r_2} \begin{bmatrix} 1 & -2 & 3 & 4 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 1 & 3 \end{bmatrix} \xrightarrow{2r_2+r_1\to r_1} \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

$$\xrightarrow[-r_3+r_1\to r_1]{r_3+r_2\to r_2} \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$ } Satırca md. basamak

α

$$A = \begin{bmatrix} 2 & 6 \\ 0 & -2 \\ -1 & 1 \\ 2 & 1 \end{bmatrix}_{4\times 2}$$

A'yı mdrgeyim.

$$\begin{bmatrix} 2 & 6 \\ 0 & -2 \\ -1 & 1 \\ 2 & 1 \end{bmatrix} \xrightarrow[-1r_2\to r_2]{\frac{1}{2}r_1\to r_1} \begin{bmatrix} 1 & 3 \\ 0 & 1 \\ -1 & 1 \\ 2 & 1 \end{bmatrix} \xrightarrow{r_1+r_3\to r_3 \\ -2r_1+r_4\to r_4} \begin{bmatrix} 1 & 3 \\ 0 & 1 \\ 0 & 4 \\ 0 & -5 \end{bmatrix} \xrightarrow{\frac{1}{4}r_3\to r_3 \\ -r_2+r_3\to r_3 \\ 5r_2+r_4\to r_4 \\ -3r_2+r_1\to r_1} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

1. adım, 2. adım

α

$$A = \begin{bmatrix} -1 & 2 & 0 & 5 \\ 1 & 4 & -4 & 1 \\ 1 & -4 & 2 & -2 \end{bmatrix} \qquad B = \begin{bmatrix} 1 & 6 & -8 & 5 \\ 2 & 8 & -8 & 2 \\ 2 & 0 & -2 & 1 \end{bmatrix}$$

- A ve B denk matrisler mi? ✓
- A ve B'nin mdrgemiş formları eşit midir? ✓

$$A=\begin{bmatrix}-1 & 2 & 0 & 5\\ 1 & 4 & -4 & 1\\ 1 & -4 & 2 & -2\end{bmatrix} \xrightarrow{r_2+r_3\to r_3} \begin{bmatrix}-1 & 2 & 0 & 3\\ 1 & 4 & -4 & 1\\ 2 & 0 & -2 & -1\end{bmatrix} \xrightarrow[-r_2+r_1\to r_1]{2r_2\to r_2} \begin{bmatrix}1 & 10 & -8 & 5\\ 2 & 8 & -8 & 2\\ 2 & 0 & -2 & -1\end{bmatrix} = B$$

∝ Kendileri sıfır olmayan ancak çarpımları sıfır matris olan iki matris bul. (sıfır bölen)

$$A_{p\times q}\neq O_{p\times q},\quad B_{q\times r}\neq O_{q\times r} \qquad \underline{AB=O_{q\times r}}$$

$$\begin{bmatrix}1 & 0\\ 0 & 0\end{bmatrix}\cdot\begin{bmatrix}0 & 0\\ 1 & 0\end{bmatrix}=\begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}_{2\times 2},\qquad \begin{bmatrix}1 & 0\end{bmatrix}_{1\times 2}\begin{bmatrix}0\\ 1\end{bmatrix}_{2\times 1}=\begin{bmatrix}0\end{bmatrix}_{1\times 1}$$

∝
$$A=\begin{bmatrix}1 & 2 & 3\\ 2 & 4 & 5\\ a & b & c\end{bmatrix} \quad \text{ve} \quad A^{-1}=\begin{bmatrix}1 & 1 & -2\\ -3 & 1 & 1\\ 2 & -1 & 0\end{bmatrix} \quad \text{ise} \quad a+b+c=?$$

$$\begin{bmatrix}1 & 1 & -2\\ -3 & 1 & 1\\ 2 & -1 & 0\end{bmatrix}\cdot\begin{bmatrix}1 & 2 & 3\\ 2 & 4 & 5\\ a & b & c\end{bmatrix}=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$$

$$A^{-1}\cdot A = I \quad \text{?} \quad A.A^{-1}=A$$

$c_{31}=2-2+0=0$

$c_{21}=-3+2+a=-1+a=0 \longrightarrow a=1$

$c_{22}=-6+4+b=1 \to b=3$

$c_{23}=-9+5+c=0 \to c=4$

$a+b+c=8$

Determinant Vize

$A = [a_{ij}]$ $n \times n$ lik matris olsun. Determinant fonksiyonu;

$\det(A) = \sum(\pm)\, a_{1p_1}, a_{2p_2} \cdots a_{np_n}$ olarak tanımlıdır. $a_{1p_1} \cdots, a_{2p_2}$ a_{np_n} tüm permütasyonları ifade eder ve $n!$ tanedir. $(\pm)$ sıralamasının (permütasyonun) tek veya çift olmasına göre belirlenir. (çift ise $+$, tek ise $-$)

- 3 elemanlı permütasyon

$$\begin{pmatrix}1&2&3\\1&2&3\end{pmatrix} \begin{pmatrix}1&2&3\\2&3&1\end{pmatrix} \begin{pmatrix}1&2&3\\3&1&2\end{pmatrix}$$

$3! = 6$

$$\begin{pmatrix}1&2&3\\1&3&2\end{pmatrix} \begin{pmatrix}1&2&3\\3&2&1\end{pmatrix} \begin{pmatrix}1&2&3\\2&1&3\end{pmatrix}$$

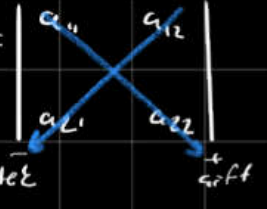
- $n=1$ için: $A = [a_{11}]$ ise $\det(A) = |A| = a_{11}$

- $n=2$ için: $A = \begin{bmatrix}a_{11}&a_{12}\\a_{21}&a_{22}\end{bmatrix}$ perm. $2! = 2$ tane

$$\begin{pmatrix}1&2\\1&2\end{pmatrix} \text{ ve } \begin{pmatrix}1&2\\2&1\end{pmatrix}$$

çift (+) tek (−)

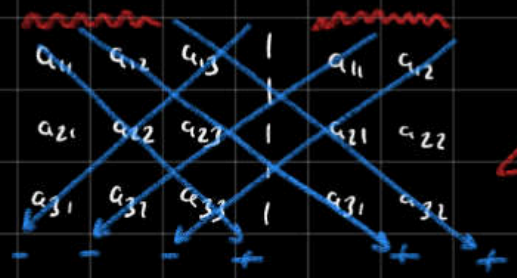
$n=2$ için determinant: $\sum + (a_{11}a_{22}) - (a_{12}a_{21})$ $|A| = \begin{vmatrix}a_{11}&a_{12}\\a_{21}&a_{22}\end{vmatrix}$



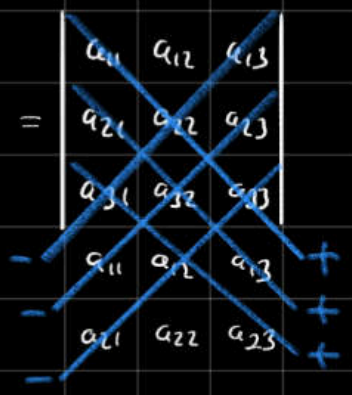
- $n=3$ için: $A = \begin{bmatrix}a_{11}&a_{12}&a_{13}\\a_{21}&a_{22}&a_{23}\\a_{31}&a_{32}&a_{33}\end{bmatrix}$ $3! = 6$ tane

$$|A| = \sum_{+}(a_{11}a_{22}a_{33}) + (a_{12}a_{23}a_{31}) + (a_{13}a_{21}a_{32}) - (a_{13}a_{22}a_{31}) - (a_{11}a_{23}a_{32}) - (a_{12}a_{21}a_{33})$$

$|A| =$



$=$



Determinantın Özellikleri:

1. $\det(A) = \det(A^t)$

2. A matrisinin iki satırının (sütunun) değişmesi halinde $(r_i \leftrightarrow r_j)$ determinant -1 ile çarpılır.

3. İki satırı (sütunu) eşit bir matrisin determinantı sıfırdır.

4. A matrisi 0 satırı içeriyorsa determinantı sıfırdır.

5. A matrisine (kri) temel işlemi uygulanırsa determinant sonucu k ile çarpılır.
   satırla çarpmada

6. A matrisine $kr_i + r_j$ temel işlemi uygulanırsa determinant sonucu değişmez.
   eklemede

7. A matrisi köşegen ise
   alt üçgensel ise
   üst üçgensel ise determinant sonucu $\det(A) = a_{11} \cdot a_{22} \cdots a_{nn}$ dir.
   (köşegendeki elemanların çarpımı)

8. $\det(AB) = \det(A)\det(B)$

9. "A^{-1} vardır" $\Leftrightarrow$ "$\det(A) \neq 0$"

10. A^{-1} varsa $\det(A^{-1}) = \dfrac{1}{\det(A)}$ dir.

satır 3.

11. A matrisi k ile çarpılırsa determinant k^n ile çarpılır.

12. $\det(A^n) = (\det A)^n$

∝ $\begin{vmatrix} 2 & -3 \\ 4 & 5 \end{vmatrix} = ?$

$-(-12)$ $+(10)$

$-(-12) + 10 = +22$

13. $\begin{vmatrix} a+x & b & c \\ d+y & e & f \\ g+z & h & 1 \end{vmatrix} = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & 1 \end{vmatrix} + \begin{vmatrix} x & b & c \\ y & e & f \\ z & h & 1 \end{vmatrix}$?

∝ $\begin{vmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 1 & 2 \end{vmatrix} = ?$

$-(9+3+8) + (2+18+6) = 8$

∝ $|A| = \begin{vmatrix} x & y \\ 1 & 4 \end{vmatrix} = 3$ $\quad B = \begin{vmatrix} 2x & 6y \\ 1 & 12 \end{vmatrix} = ?$

$4x - y = 3$ $\quad 24x - 6y = 6(4x - y) = 6 \cdot 3 = 18$

∝ $A = \begin{bmatrix} 4 & 3 & 2 \\ 3 & -2 & 5 \\ 2 & 4 & 6 \end{bmatrix}$ A'yı basamak matris yaparak determinant hesaplayın.

$\begin{bmatrix} 4 & 3 & 2 \\ 3 & -2 & 5 \\ 2 & 4 & 6 \end{bmatrix} \rightarrow |A| 2 \begin{vmatrix} 4 & 3 & 2 \\ 3 & -2 & 5 \\ 1 & 2 & 3 \end{vmatrix} \xrightarrow[\substack{-3r_3+r_2 \to r_2 \\ -4r_3+r_1 \to r_1}]{r_3 \leftrightarrow r_1} \begin{vmatrix} 0 & 5 & -10 \\ 0 & -8 & -4 \\ 1 & 2 & 3 \end{vmatrix} \xrightarrow[\substack{-\frac{1}{5}r_1 \to r_1 \\ -\frac{1}{4}r_2 \to r_2}]{(-2)(-4)(-5)} \begin{vmatrix} 0 & 1 & 2 \\ 0 & 2 & 1 \\ 1 & 2 & 3 \end{vmatrix}$

$$\begin{vmatrix} 0 & 1 & 2 \\ 0 & 2 & 1 \\ 1 & 2 & 3 \end{vmatrix} \xrightarrow[-2r_1+r_2 \to r_2]{40.} \begin{vmatrix} 0 & 1 & 2 \\ 0 & 0 & -3 \\ 1 & 2 & 3 \end{vmatrix} \xrightarrow[r_1 \leftrightarrow r_2]{40.} \begin{vmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & -3 \end{vmatrix} = 40(-3) = -120$$

$-(0) \qquad +(-3)$

α

$$\begin{vmatrix} a & a & a & a \\ b & b & b & b \\ c & c & c & c \\ d & d & d & d \end{vmatrix} = a.b.c.d \begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{vmatrix} \xrightarrow[-r_1+r_3 \to r_3]{-r_1+r_2 \to r_2} a.b.c.d \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & -2 \\ 0 & 0 & -2 & -2 \\ 0 & -2 & -2 & -2 \end{vmatrix}$$

$-r_1+r_4 \to r_4$

$$\xrightarrow{r_2 \leftrightarrow r_4} -a.b.c.d \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & -2 & -2 & -2 \\ 0 & 0 & -2 & -2 \\ 0 & 0 & 0 & -2 \end{vmatrix} = -a.b.c.d.(-2)(-2)(-2)$$

Üst üçgensel $\qquad +(-2)(-2)(-2)$

α

$$\begin{vmatrix} 1 & 4 & 1 & -4x \\ 3 & -1 & 3 & x \\ 2 & -1 & 7 & x \\ 5 & 1 & 1 & -x \end{vmatrix} \xrightarrow{-x} \begin{vmatrix} 1 & 4 & 1 & 4 \\ 3 & -1 & 3 & -1 \\ 2 & -1 & 7 & -1 \\ 5 & 1 & 1 & 1 \end{vmatrix} \xrightarrow[-x]{-c_2+c_4 \to c_4} \begin{vmatrix} 1 & 4 & 1 & 0 \\ 3 & -1 & 3 & 0 \\ 2 & -1 & 3 & 0 \\ 5 & 1 & 1 & 0 \end{vmatrix} = 0$$

4x4 $\qquad$ 4 0 ?

Sütunları lineer bağımlı matris
Tekil matris, ters (A^{-1}) yok

α

$$\begin{vmatrix} x & a & 2ax \\ a & 3 & a^2+3x \\ 1 & 1 & a+x \end{vmatrix} = ?$$

$$\begin{vmatrix} x & a & 2ax \\ a & 3 & a^2+3x \\ 1 & 1 & a+x \end{vmatrix} \xrightarrow{-ac_1+c_3 \to c_3} \begin{vmatrix} x & a & ax \\ a & 3 & 3x \\ 1 & 1 & x \end{vmatrix} \xrightarrow{-xc_2+c_3 \to c_3} \begin{vmatrix} x & a & 0 \\ a & 3 & 0 \\ 1 & 1 & 0 \end{vmatrix} = 0$$

α $\begin{vmatrix} x+1 & x+1 & 1 \\ -2x-3 & -x & -3 \\ 1 & 0 & 1 \end{vmatrix}_{3\times 3} = 3$ ise $x = ?$

$$\begin{vmatrix} x+1 & x+1 & 1 \\ -2x-3 & -x & -3 \\ 1 & 0 & 1 \end{vmatrix} \xrightarrow[3r_3+r_2\to r_2]{-r_3+r_1\to r_1} \begin{vmatrix} x & x+1 & 0 \\ -2x & -x & 0 \\ 1 & 0 & 1 \end{vmatrix} \xrightarrow[-c_3+c_1\to c_1]{2r_1+r_2\to r_2} \begin{vmatrix} x & x+1 & 0 \\ 0 & x+2 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 3$$

$$x(x+2) = x^2+2x = 3 \rightarrow x^2+2x-3 = (x+3)(x-1)$$
$$x \quad +3$$
$$x \quad -1$$
$x_1 = -3,\ x_2 = 1$

α $\begin{vmatrix} 1 & 0 & 3 & 0 \\ 0 & 3 & 0 & 1 \\ 3 & 0 & 1 & 0 \\ 0 & 1 & 0 & 3 \end{vmatrix} \xrightarrow[-3r_1+r_2\to r_2]{-3r_1+r_3\to r_3} \begin{vmatrix} 1 & 0 & 3 & 0 \\ 0 & 0 & 0 & 8 \\ 0 & 0 & -8 & 0 \\ 0 & 1 & 0 & 3 \end{vmatrix} \xrightarrow{r_2 \Leftrightarrow r_4} \begin{vmatrix} 1 & 0 & 3 & 0 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & -8 & 0 \\ 0 & 0 & 0 & -8 \end{vmatrix} = -64$

α $\begin{vmatrix} 1 & a & 2 \\ -1 & 1 & b \\ a & 2 & 3b \end{vmatrix} = 4$ ise;

a) $\begin{vmatrix} x^2 & ax & 2x \\ -x & 1 & b \\ ax & 2 & 3b \end{vmatrix} = ? = xx \begin{vmatrix} 1 & a & 2 \\ -1 & 1 & b \\ a & 2 & 3b \end{vmatrix} = 4x^2$

x parantez

b) $\begin{vmatrix} x & a+bx & 2 \\ -x & 1-bx & b \\ ax & 2+abx & 3b \end{vmatrix}_{3\times 3} = ? = \begin{vmatrix} 1 & a & 2 \\ -1 & 1 & b \\ a & 2 & 3b \end{vmatrix} = 4x$

x +

c) $\begin{vmatrix} -1 & -2a & -2 \\ 2 & 4 & 2b \\ a & 4 & 3b \end{vmatrix}$ $\begin{matrix} -1 \\ 2 \end{matrix}$ $= ? = 2 \cdot 2 \cdot -1 \begin{vmatrix} 1 & a & 2 \\ -1 & 1 & b \\ a & 2 & 3b \end{vmatrix} = -16$

d) $\begin{vmatrix} a+1 & a+2 & 3b+2 \\ -1 & 1 & b \\ a & 2 & 3b \end{vmatrix}_{3\times 3} = ?$

Minör ve Kofaktör

$A = [a_{ij}]_{n\times n}$ lik kare matris olsun A'nın ij minörü $\det(M_{ij})$ dir. M_{ij} ise A matrisinin i satır ve j sütununun silinmesiyle elde edilir.

α $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \longrightarrow M_{12} = \begin{bmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{bmatrix}$ Minor $\rightarrow \det(M_{12}) = \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}$

- A_{ij} kofaktör ise $A_{ij} = (-1)^{i+j} \cdot \det(M_{ij})$ olarak tanımlıdır.

α $A_{12} = (-1)^{1+2} \cdot \underbrace{\det(M_{12})}_{\text{Minor}}$ (M_{12}: alt matris)

α $A = \begin{bmatrix} 3 & -1 & 2 \\ 4 & 5 & 6 \\ 7 & 1 & 2 \end{bmatrix}$ için;

$A_{12} = (-1)^{1+2} \cdot \underbrace{\det(M_{12})}_{-34}$ $M_{12} = \begin{bmatrix} 4 & 6 \\ 7 & 2 \end{bmatrix}$ $\begin{matrix} -42 & +8 \end{matrix}$ $A_{12} = -1 \cdot -34 = 34$

$A_{23} = (-1)^{2+3} \cdot \underbrace{\det(M_{23})}_{10}$ $\quad M_{23} = \begin{bmatrix} 3 & -1 \\ 7 & 1 \end{bmatrix}$ $\quad A_{23} = -1 \cdot 10 = \underline{-10}$

$-(-1) \qquad 3$

$A_{31} = (-1)^{3+1} \cdot \underbrace{\det(M_{31})}_{-16}$ $\quad M_{31} = \begin{bmatrix} -1 & 2 \\ 5 & 6 \end{bmatrix}$ $\quad A_{31} = 1 \cdot -16 = \underline{-16}$

$-10 \qquad -6$

Genel Determinant

$A = [a_{ij}]_{n \times n}$ için $\det(A) = a_{i1}A_{i1} + a_{i2}A_{i2} + \cdots + a_{in}A_{in}$

$\det(A) = a_{1j}A_{1j} + a_{2j}A_{2j} + \cdots + a_{nj}A_{nj}$

olarak tanımlıdır.

örn $A = \begin{bmatrix} -2 & -1 & 0 \\ -1 & 2 & 1 \\ 0 & 3 & -2 \end{bmatrix}$ $\quad |A| = ?$

Satırda

$|A| = 0 \cdot A_{31} + 3 \cdot A_{32} + (-2) \cdot A_{33}$

$= 3 \cdot (-1)^{3+2} \cdot \det(M_{32}) + (-2)(-1)^{3+3} \cdot \det(M_{33})$

$= -3 \begin{vmatrix} -2 & 0 \\ -1 & 1 \end{vmatrix} - 2 \begin{vmatrix} -2 & -1 \\ -1 & 2 \end{vmatrix} = 6 + 10 = \underline{16}$

$0 \qquad -2 \qquad -1 \qquad -4$

$-2 \qquad -5$

Sütunca

$|A| = 0 \cdot A_{13} + 1 \cdot A_{23} + (-2)A_{33} = 1(-1)^{2+3} \det(M_{23}) + (-2)(-1)^{3+3} \cdot \det(M_{33})$

$= - \begin{vmatrix} -2 & -1 \\ 0 & 3 \end{vmatrix} - 2 \begin{vmatrix} -2 & -1 \\ -1 & 2 \end{vmatrix} = 6 + 10 = \underline{16}$

$0 \quad -6 \quad -1 \quad -4$

$-6 \qquad -5$

$$\alpha A = \begin{vmatrix} 1 & 2 & -3 & 4 \\ -4 & 2 & 1 & 3 \\ 3 & 0 & 0 & -3 \\ 2 & 0 & -2 & 3 \end{vmatrix} = ? \xrightarrow{-r_1+r_2 \to r_2} \begin{vmatrix} 1 & 2 & -3 & 4 \\ -5 & 0 & 4 & -1 \\ 3 & 0 & 0 & -3 \\ 2 & 0 & -2 & 3 \end{vmatrix} = 2 \cdot A_{12} + 0 \cdot \cancel{A_{22}}^{0} + 0 \cdot \cancel{A_{32}}^{0} + 0 \cdot \cancel{A_{42}}^{0}$$

$$= 2A_{12} = 2 \cdot (-1)^{1+2} \det(M_{12}) = -2 \underbrace{\begin{vmatrix} -3 & 4 & -1 \\ 3 & 0 & -3 \\ 2 & -2 & 3 \end{vmatrix}}_{13} \xrightarrow{c_1+c_3 \to c_3} \begin{vmatrix} -5 & 4 & -6 \\ 3 & 0 & 0 \\ 2 & -2 & 5 \end{vmatrix}$$

$$= 3B_{21} + 0 \cdot \cancel{B_{22}}^{0} + 0 \cdot \cancel{B_{23}}^{0}$$

$$= -3(-1)^{2+1} \det(M_{12}) = -3 \begin{vmatrix} 4 & -6 \\ -2 & 5 \end{vmatrix}$$

$$= -24$$

$$-2B = -2 \cdot -24 = 48$$

$$\alpha \begin{vmatrix} 1 & 2 & -3 & 4 \\ -4 & 2 & 1 & 3 \\ 1 & 0 & 0 & -3 \\ 2 & 0 & -2 & 3 \end{vmatrix} = ? \xrightarrow{-3c_1+c_4 \to c_4} \begin{vmatrix} 1 & 2 & -3 & 1 \\ -9 & 2 & 1 & 9 \\ 1 & 0 & 0 & 0 \\ 2 & 0 & -2 & -3 \end{vmatrix}$$

$$1 \cdot A_{31} = \begin{vmatrix} 2 & -3 & 1 \\ 2 & 1 & 9 \\ 0 & -2 & -3 \end{vmatrix} \xrightarrow{-r_1+r_2 \to r_2} \begin{vmatrix} 2 & -3 & 1 \\ 0 & 4 & 8 \\ 0 & -2 & -3 \end{vmatrix} \quad 2 \cdot A_{11} = 2 \underbrace{\begin{vmatrix} 4 & 8 \\ -2 & -3 \end{vmatrix}}_{-(-16) \quad -12} = 4 \cdot 2 = 8$$

4

$$\alpha\ |A| = \begin{vmatrix} 1 & 1 & -1 & 2 \\ 0 & 1 & 3 & 1 \\ 1 & 2 & -1 & 0 \\ -1 & 2 & 1 & 2 \end{vmatrix} \xrightarrow[r_1+r_4 \to r_4]{-r_1+r_2 \to r_2} \begin{vmatrix} 1 & 1 & -1 & 2 \\ 0 & 1 & 3 & 1 \\ 0 & 1 & 0 & -2 \\ 0 & 3 & 0 & 4 \end{vmatrix} = 1.A_{11} + \cancel{0.A_{21}} + \cancel{0.A_{31}} + \cancel{0.A_{41}}$$

$$= 1.(-1)^{1+1} \begin{vmatrix} 1 & 3 & 1 \\ 1 & 0 & -2 \\ 3 & 0 & 4 \end{vmatrix}$$

$$= 3.A_{21} + 0.A_{22} + 0.A_{32}$$

$$= -3. \begin{vmatrix} 1 & -2 \\ 3 & 4 \end{vmatrix} = -3(4+6) = \underline{-30}$$

Ek Matris Vize

$A = [a_{ij}]_{n \times n}$ matris olmak üzere $Ek(A) = Adj(A)$ matrisi A matrisinin kofaktörleri A_{ij} ile $\underbrace{Ek(A)}_{\text{Ek}} = [A_{ji}] = [A_{ij}]^t$ olarak yazılır.
kofaktör

α $A = \begin{bmatrix} 3 & -2 & 1 \\ 5 & 6 & 2 \\ 1 & 0 & -3 \end{bmatrix}$ matrisi için $Ek(A) = ?$

$$Ek(A) = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}^t = \begin{bmatrix} A_{11} & A_{21} & A_{31} \\ A_{12} & A_{22} & A_{32} \\ A_{13} & A_{23} & A_{33} \end{bmatrix}$$

$Ek(A) = [A_{ji}] = [A_{ij}]^t$

$$A_{11} = (-1)^{1+1}.\det(M_{11}) = 1. \begin{vmatrix} 6 & 2 \\ 0 & -3 \end{vmatrix} = -18 \qquad A_{13} = (-1)^4.M_{13} = \begin{vmatrix} 5 & 6 \\ 1 & 0 \end{vmatrix} = -6$$

$$A_{12} = (-1)^{1+2}.\det(M_{12}) = -1. \begin{vmatrix} 5 & 2 \\ 1 & -3 \end{vmatrix} = 17$$

$$A_{21} = (-1)^{2+1} \cdot \det(M_{21}) = -1 \cdot \begin{vmatrix} -2 & -1 \\ 0 & 3 \end{vmatrix} = +6$$

$$Ek(A) = \begin{bmatrix} -18 & 17 & -6 \\ 6 & -10 & -2 \\ -10 & -1 & 28 \end{bmatrix}^t = \begin{bmatrix} -18 & 6 & -10 \\ 17 & -10 & -1 \\ -6 & -2 & 28 \end{bmatrix}$$

trampoz

Ek Matris ile Matrisin Tersi: $A_{n\times n}$ ve $|A| \neq 0$ ise

Vize

$$A^{-1} = \frac{1}{\det(A)} \cdot Ek(A) = \frac{Ek(A)}{|A|}$$

α $A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & -1 & 3 \\ 4 & 1 & 8 \end{bmatrix}$ A matrisinin tersi $A^{-1} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$ ise d uch?

$$A_{11} = (-1)^{1+1} \cdot \det(M_{11}) = 1 \cdot \begin{vmatrix} -1 & 3 \\ 1 & 8 \end{vmatrix} = -11$$

$$A_{12} = (-1)^{1+2} \cdot \det(M_{12}) = -1 \cdot \begin{vmatrix} 2 & 3 \\ 4 & 8 \end{vmatrix} = -4$$

$-(16)$ 16

İndirgeme

$$[A_{ij}] = \begin{bmatrix} -11 & -4 & 6 \\ 2 & 0 & -1 \\ 2 & 1 & -1 \end{bmatrix} \xrightarrow{t} Ek(A) = \begin{bmatrix} -11 & 2 & 2 \\ -4 & 0 & 1 \\ 6 & -1 & -1 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 1 & 0 & 2 \\ 2 & -1 & 3 \\ 4 & 1 & 8 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 2 \\ 0 & -1 & -1 \\ 0 & 1 & 0 \end{vmatrix} = 1$$

$$A^{-1} = \frac{Ek(A)}{1} = Ek(A)$$

α $A = \begin{bmatrix} 1 & -2 & -1 \\ 3 & -1 & 4 \\ -2 & -3 & 2 \end{bmatrix}$ matrisin tersinin ilk satırını bulunuz.

matrisin determinantının ilk satırı

$A_{11} = (-1)^{1+1} \cdot \det(M_{11}) = 1 \cdot \begin{vmatrix} -1 & 4 \\ -3 & 2 \end{vmatrix} = 10$

$A_{21} = (-1)^{2+1} \cdot \begin{vmatrix} -2 & -1 \\ -3 & 2 \end{vmatrix} = +7$

$A_{31} = (-1)^{4} \cdot \begin{vmatrix} -2 & -1 \\ -1 & 4 \end{vmatrix} = -9$

$\begin{bmatrix} 10 & \cdot & \cdot \\ 7 & \cdot & \cdot \\ -9 & \cdot & \cdot \end{bmatrix}^t = \begin{bmatrix} 10 & 7 & -9 \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{bmatrix}$ $Ek(A)$

$|A| = \begin{vmatrix} 1 & -2 & -1 \\ 3 & -1 & 4 \\ -2 & -3 & 2 \end{vmatrix} = 1 \cdot \underbrace{A_{11}}_{10} + (-2)\underbrace{A_{12}}_{-14} + (-1)\underbrace{A_{13}}_{-11} = 49$

$A^{-1} = \frac{1}{\det(A)} \cdot Ek(A) = \begin{bmatrix} \frac{10}{49} & \frac{7}{49} & \frac{-9}{49} \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{bmatrix}_{3\times3}$

α $A = \begin{bmatrix} 1 & 0 \\ 2 & -1 \\ 5 & 3 \end{bmatrix}_{3\times2}$ $B = \begin{bmatrix} -1 \\ 2 \end{bmatrix}_{2\times1}$ $A \cdot B = ?$

$\begin{bmatrix} 1 & 0 \\ 2 & -1 \\ 5 & 3 \end{bmatrix}_{3\times2} \cdot \begin{bmatrix} -1 \\ 2 \end{bmatrix}_{2\times1} = \begin{bmatrix} -1+0=-1 \\ -2-2=-4 \\ -5+6=1 \end{bmatrix}_{3\times1}$

α $A = \begin{bmatrix} -1 & 0 & -1 & 2 & 1 \\ 2 & -1 & 0 & 1 & 0 \\ 1 & -1 & -1 & 0 & 1 \end{bmatrix}_{3\times5}$ A'yı indirgeyin. → hep satırca

$$\begin{bmatrix} -1 & 0 & -1 & 2 & 1 \\ 2 & -1 & 0 & 1 & 0 \\ 1 & -1 & -1 & 0 & 1 \end{bmatrix} \xrightarrow[r_1+r_3\to r_3]{2r_1+r_2\to r_2} \begin{bmatrix} -1 & 0 & -1 & 2 & 1 \\ 0 & -1 & -2 & 5 & 2 \\ 0 & -1 & -2 & 2 & 2 \end{bmatrix} \xrightarrow[-r_2+r_3\to r_3]{-r_1\to r_1} \begin{bmatrix} 1 & 0 & 1 & -2 & -1 \\ 0 & -1 & -2 & 5 & 2 \\ 0 & 0 & 0 & -3 & 0 \end{bmatrix}$$

$$\xrightarrow[\frac{1}{3}r_3\to r_3]{-r_2\to r_2} \begin{bmatrix} 1 & 0 & 1 & -2 & -1 \\ 0 & 1 & 2 & -5 & -2 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} \xrightarrow[5r_3+r_2\to r_2]{2r_3+r_1\to r_1} \begin{bmatrix} 1 & 0 & 1 & 0 & -1 \\ 0 & 1 & 2 & 0 & -2 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}_{3\times5}$$

α $A^{-1} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & -1 & 1 \end{bmatrix}$ ise $A = ?$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 2 & 0 & 1 & 0 \\ 1 & -1 & 1 & 0 & 0 & 1 \end{array}\right] \xrightarrow[-r_1+r_3\to r_3]{-r_1+r_2\to r_2} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 1 & 0 \\ 0 & -2 & 0 & -1 & 0 & 1 \end{array}\right]$$

$[A^{-1} \mid I]$

$$\xrightarrow{\frac{1}{2}r_3+r_1\to r_1} \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1/2 & 0 & 1/2 \\ 0 & 0 & 1 & -1 & 1 & 1 \\ 0 & -2 & 0 & -1 & 0 & 1 \end{array}\right] \xrightarrow[-\frac{1}{2}r_3\to r_3]{-r_2+r_1\to r_1} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3/2 & -1 & 1/2 \\ 0 & 0 & 1 & -1 & 1 & 0 \\ 0 & 1 & 0 & 1/2 & 0 & -1/2 \end{array}\right]$$

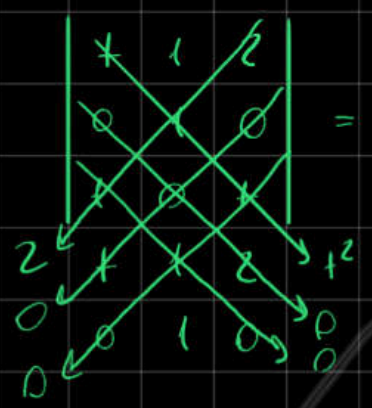
$$\longrightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3/2 & -1 & 1/2 \\ 0 & 1 & 0 & 1/2 & 0 & -1/2 \\ 0 & 0 & 1 & -1 & 1 & 0 \end{array}\right]$$

$[I \mid \underbrace{((A^{-1})^{-1})}_{A}]$

$\alpha \begin{bmatrix} t & 1 & 2 \\ 0 & 1 & 0 \\ 1 & 0 & t \end{bmatrix}$ A^{-1} 'in var olması için $t = ?$

$A^{-1} = \frac{1}{\det(A)} \cdot Ek(A)$ $\quad \det(A) \neq 0$

$\begin{vmatrix} t & 1 & 2 \\ 0 & 1 & 0 \\ 1 & 0 & t \end{vmatrix} =$



$= t^2 - 2 \neq 0$

$t^2 \neq 2$

$t \neq \sqrt{2}$

$t \neq -\sqrt{2}$

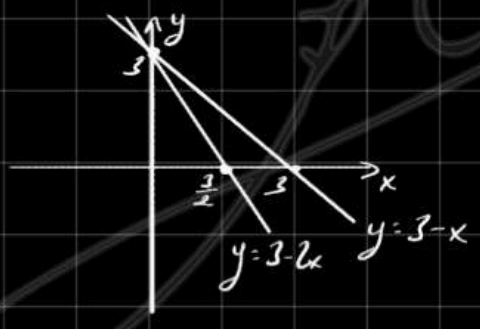
Lineer Denklem Sistemlerine Giriş

($x^2, x.y$ vs. olursa olmaz)

1) $2x + y = 3$
$x + y = 3$

İncer denklem sisteminin çözümünü inceleyim.

$y = 3 - 2x$
$y = 3 - x$

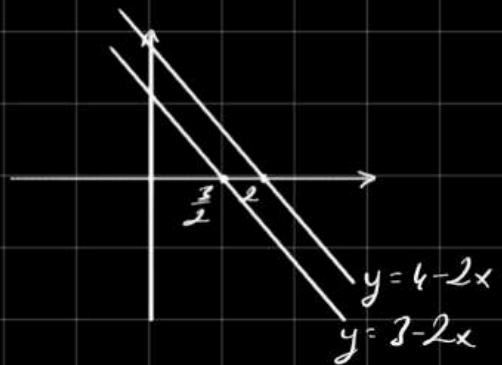


$d_1 \cap d_2 = \{(0,3)\}$
$Ç.K. = \{(0,3)\}$

2) $2x + y = 4$
$4x + 2y = 6$

Sisteminin çöz. inceleyim.

$y = 4 - 2x$
$y = 3 - 2x$

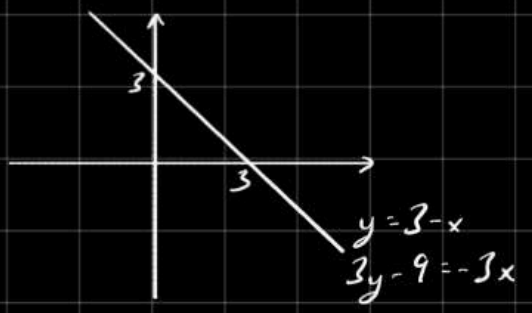


$d_1 \parallel d_2$
$d_1 \cap d_2 = \emptyset$
$Ç.K. = \emptyset$

3) $3-x=y$
$3y-9=-3x$

sistemnin çöz. inceleyim.

$y=3-x$
$y=3-x$



$d_1 \cap d_2 = d_1 = d_2$
sonsuz noktada verdir.
$\subseteq K. = \{(x, 3-x) \mid x \in R\}$
parametreye bağlı sonsuz çözüm

Lineer Denklem Sistemi Vize

Tanım:
$b, a_1, a_2, \cdots a_n \in R$
$x_1, x_2, x_3, \cdots x_n$ bilinmeyenler olmak üzere $a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$
denklemine n bilinmeyenli bir lineer denklem denir.

a_i'ler denklemin katsayıları
b ise denklemin sabitidir.

Tanım:

$$a_{11}x_1 + a_{12}x_2 + \cdots a_{1n}x_n = b_1$$
$$a_{21}x_1 + a_{22}x_2 + \cdots a_{2n}x_n = b_2$$
$$\vdots$$
$$a_{p1}x_1 + a_{p2}x_2 + \cdots a_{pn}x_n = b_p$$

şeklinde p lineer denklemden oluşan yapıya n bilinmeyenli p tane denklemden oluşan lineer denklem sistemi denir.

Tanım: Lineer denklem sistemini sağlayan değerler bütününe Ç.K. denir.

Tanım: Lineer denklem sisteminin denklem sabitleri b_i'ler $b_1 = b_2 = b_n = 0$ ise bu tür sistemlere homojen lineer denklem sistemi denir.

Tanım: Lineer denklem sistemlerinin Ç.K.'sı için 3 farklı durumdan birisi geçerlidir;

1) Sistemin çözümü yoktur.

Tüm denk. sağlayan en az bir durum bulunmadığı durumlar.

2) Sistemin bir tek çözümü vardır.

Tüm denk. sağlayan yalnız bir çözüm vardır.

3) Sistemin sonsuz çözümü vardır.

Bir veya daha fazla değişkene (parametreye) bağlı çözüm bulunduğu durumlar.

Tanım: Bir lineer denk. sis. matris gösterimi katsayılar matrisi A, değişkenler matrisi X ve sabitler matrisi B olmak üzere aşağıdaki gibidir.

$$\left.\begin{array}{l} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ \vdots \\ a_{p1}x_1 + a_{p2}x_2 + \cdots + a_{pn}x_n = b_p \end{array}\right\} \quad \underbrace{\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \\ a_{p1} & a_{p2} & & a_{pn} \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}}_{X} = \underbrace{\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_p \end{bmatrix}}_{B}$$

Not: Lineer denk. sis. çözülürken kolaylık olması için $[A \,|\, B]$ matrisi oluşturulur. İndirgemelerden sonra değişkenlikteki göre işlem yapılır.

$$\left[\begin{array}{ccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ \vdots & & & & \vdots \\ a_{p1} & a_{p2} & \cdots & a_{pn} & b_n \end{array}\right] \longrightarrow \left[\begin{array}{c|c} A' & b' \end{array}\right]$$

- $x_1 + x_2 + x_3 = 2$
  $x_2 + x_3 = 3$ $\quad C.K = ?$
  $x_1 - x_2 - x_3 = -3$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & 1 & 3 \\ 1 & -1 & -1 & -3 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 1 & 3 \\ 1 & 0 & 0 & 0 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 0 & 1 \end{array}\right]$$

$$\longrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1 \\ 3 \\ 1 \end{bmatrix} \longrightarrow x_1 = -1$$

$x_2 + x_3 = 3 \quad 0 \neq 1 \quad C.K. = \emptyset$

- $x_1 + x_2 + 2x_3 + x_5 = 2$ $\quad C.K. = ?$
  $x_4 + 2x_5 = 1$

$$\begin{bmatrix} 1 & 1 & 2 & 0 & 1 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Parametre P.

$$\left[\begin{array}{ccccc|c} 1 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 & 2 & 1 \end{array}\right]$$

$x_1 + x_2 + 2x_3 + x_5 = 2$

$x_1 = 2 - x_2 - 2x_3 - x_5$

$x_4 = 1 - 2x_5$

$$C.K. = \{(2 - x_2 - 2x_3 - x_5, x_2, x_3, 1 - 2x_5, x_5) \mid x_2, x_3, x_5 \in R\}$$

- $x + 2y + 3z = 9$
  $-y + 2x + z = 8$ $\quad C.K = ?$
  $-z + 3x = 3$

1. x y z

$$x+2y+3z=9$$
$$2x-y+z=8$$
$$3x-z=3$$

2. $A \cdot X = B$

$$\begin{bmatrix}1&2&3\\2&-1&1\\3&0&-1\end{bmatrix}\begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}9\\8\\3\end{bmatrix}$$

3.

$$\left[\begin{array}{ccc|c}1&2&3&9\\2&-1&1&8\\3&0&-1&3\end{array}\right] \rightarrow \left[\begin{array}{ccc|c}1&2&3&9\\0&-5&-5&-10\\0&-6&-10&-24\end{array}\right] \rightarrow \left[\begin{array}{ccc|c}1&2&3&9\\0&1&1&2\\0&3&5&12\end{array}\right] \rightarrow \left[\begin{array}{ccc|c}1&0&1&5\\0&1&1&2\\0&0&2&6\end{array}\right]$$

4.

$$\rightarrow \left[\begin{array}{ccc|c}1&0&1&5\\0&1&1&2\\0&0&1&3\end{array}\right] \rightarrow \left[\begin{array}{ccc|c}1&0&0&2\\0&1&0&-1\\0&0&1&3\end{array}\right] \rightarrow \left[\begin{array}{ccc|c}1&0&0&x\\0&1&0&y\\0&0&1&z\end{array}\right] = \begin{bmatrix}2\\-1\\3\end{bmatrix}$$

5. $\subseteq K = \{(2,-1,3)\}$

- $x+2y-3z=-4$
  $2x+y-3z=4$ $\quad \subseteq K = ?$

1.

$$\begin{bmatrix}1&2&-3\\2&1&-3\end{bmatrix}\begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}-4\\4\end{bmatrix}$$

2.

$$\left[\begin{array}{ccc|c}1&2&-3&-4\\2&1&-3&4\end{array}\right] \xrightarrow{-2r_1+r_2\rightarrow r_2} \left[\begin{array}{ccc|c}1&2&-3&-4\\0&-3&3&12\end{array}\right]$$

3. x y z

$$\xrightarrow{-\frac{1}{3}r_2\rightarrow r_2} \left[\begin{array}{ccc|c}1&2&-3&-4\\0&1&-1&-4\end{array}\right] \xrightarrow{-2r_2+r_1\rightarrow r_1} \left[\begin{array}{ccc|c}1&0&-1&4\\0&1&-1&-4\end{array}\right]$$

Pivot Parametre

4.

$$x-z=4 \quad x=4+z$$
$$y-z=-4 \quad y=z-4$$

5. $\subseteq K = \{(4+z,-4+z) \mid z \in R\}$

Rank

Bir A matrisinin rankı A matrisinin satırca indirgenmiş formunun pivot sayısına veya sıfırdan farklı satır sayısına eşittir.

- $A = \begin{bmatrix} -1 & 2 & -1 & -2 \\ 2 & 1 & 2 & -1 \\ 1 & 2 & 1 & -2 \\ 0 & 1 & 0 & -1 \end{bmatrix}$ A'nın rankı kaçtır?

$$\begin{bmatrix} -1 & 2 & -1 & -2 \\ 2 & 1 & 2 & -1 \\ 1 & 2 & 1 & -2 \\ 0 & 1 & 0 & -1 \end{bmatrix} \longrightarrow \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{Rank}(A) = 2$$

Rank ve Lineer Denklem Sayısına Göre Lineer Denklem Çözme

Lineer denklem sistemi

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ &\vdots \\ a_{p1}x_1 + a_{p2}x_2 + \cdots + a_{pn}x_n &= b_p \end{aligned}$$

olarak verilen lineer denklem sisteminin ekli matrisi

$$[A \mid b] = \left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{p1} & a_{p2} & \cdots & a_{pn} & b_n \end{array}\right] \quad \text{olsun.}$$

Değişken sayısı = n ve Rank(A) = r ise

1) r = n durumunda sistemin tek çözümü vardır.

2) n > r durumunda

a) ÇK = ∅ ya da b) n − r parametreye bağlı sonsuz çözüm vardır.

Örnek

$$x + y + z = 1$$
$$x - y + z = 0$$
$$3x + y + 3z = 2$$
$$x + y - z = 5$$

Çöz = ? rank 4 olabilir mi? X

Çözüm: $x = 5/2$, $y = 1/2$, $z = -2$

n = r tek çözüm

Rank(A) = 3

$$\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 0 & 0 \\ 3 & 1 & 3 & 2 & 2 \\ 1 & 1 & -1 & 5 & 5 \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 5/2 \\ 0 & 1 & 0 & 0 & 1/2 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 4 & 0 \end{array}\right]$$

$(r_2 + r_4 \rightarrow r_2)$, $-3r_2 + r_3 \rightarrow r_3$, $-r_3 + r_4 \rightarrow r_4$

$$\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 5 \\ 3 & 1 & 3 & 2 & 2 \\ 1 & 1 & -1 & 5 & 5 \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 5 \\ 0 & 1 & 3 & 2 & -13 \\ 1 & 1 & -1 & 5 & 5 \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 5 \\ 0 & 1 & 3 & 2 & -13 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & 5 & 5 \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 5 \\ 0 & 1 & 3 & 2 & -13 \\ 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & -2 & 4 & 0 \end{array}\right]$$

$$3x_1 + x_2 + x_3 + x_4 = 1$$
$$-2x_1 + x_2 + x_3 + x_4 = 0$$
$$x_1 + x_2 + x_3 + x_4 = -1$$

- ÇK = ?
- Tek çözüm olabilir mi?

$$\left[\begin{array}{cccc|c} 3 & 1 & 1 & 1 & 1 \\ -2 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & -1 \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1 \\ -3 & 0 & 0 & 0 & 4 \\ 0 & 0 & 1 & 0 & -2 \end{array}\right]$$

$-2r_3 + r_1 \rightarrow r_1$, $3r_3 + r_2 \rightarrow r_2$

rank = 1 ?

Sınavda Gider

α $x+y-4z=0$
$2x+3y+z=1$
$4x+7y+kz=t$

Lineer denk sis k ve $t=?$

$$\left[\begin{array}{ccc|c} 1 & 1 & -4 & 0 \\ 2 & 3 & 1 & 1 \\ 4 & 7 & k & t \end{array}\right] \xrightarrow[-4r_1+r_3\to r_3]{-2r_1+r_2\to r_2} \left[\begin{array}{ccc|c} 1 & 1 & -4 & 0 \\ 0 & 1 & 9 & 1 \\ 0 & 3 & 16+k & t \end{array}\right] \xrightarrow[-r_2+r_1\to r_1]{-3r_2+r_3\to r_3} \left[\begin{array}{ccc|c} 1 & 0 & -13 & -1 \\ 0 & 1 & 9 & 1 \\ 0 & 0 & -11+k & -3+t \end{array}\right]$$

① $k-11\neq 0$ ise $rank(A)=3$ bu durumda tek çözüm vardır.

= bilinmeyen sayısı

② $k-11=0$ ise $rank(A)=2\neq$ bilinmeyen sayısı

$t-3=0$ ise

$$\left[\begin{array}{ccc|c} 1 & 0 & -13 & -1 \\ 0 & 1 & 9 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right] \begin{array}{l} \to x-13z=-1 \to x=13z-1 \\ \to y+9z=1 \to y=1-9z \end{array}$$

$t-3\neq 0$ ise

$$\left[\begin{array}{ccc|c} 1 & 0 & -13 & -1 \\ 0 & 1 & 9 & 1 \\ 0 & 0 & 0 & t-3 \end{array}\right] \quad 0\neq 3$$

$\mathcal{G}.K=\{(-1+13z,\ 1-9z,\ z)\}$

Sonsuz çözüm

$\mathcal{G}.K=\emptyset$

Örnek

$x + ky + z = k$, $2x + 4y + kz = 4$ k ya göre incele.

$$\left[\begin{array}{ccc|c} 1 & k & 1 & k \\ 2 & 4 & k & 4 \end{array}\right] \xrightarrow{-2r_1+r_2 \to r_2} \left[\begin{array}{ccc|c} 1 & k & 1 & k \\ 0 & 4-2k & k-2 & -4-2k \end{array}\right]$$

k = 2

Rank(A) = 1

$0 \neq -8$

$\zeta K = \emptyset$

k ≠ 2

a) Rank(A) = 2

3-2 = 1 parametreli sonsuz çözüm

$$\zeta K = \left\{\left(k - 2 - ky,\ -4 - 2 + \frac{k(4-2)z}{4-2k} \;\middle|\; z \in \mathbb{R}\right)\right\}$$

$$\left[\begin{array}{ccc|c} 1 & k & 1 & k \\ 2 & 4 & k & 4 \end{array}\right]$$

Cramer Yöntemi (Sadece tek çözüm bulurken)

bilinmeyen sayısı = denklem sayısı

Anxn lik katsayılar matrisine sahip lineer denklem sistemi için

det(A) ≠ 0 ise çözüm bulunması için $A_1 = \left[\begin{array}{c|ccc} b & a_{12} & a_{13} & \cdots & a_{1n} \\ \vdots & & & & \\ & a_{n2} & a_{n3} & \cdots & a_{nn} \end{array}\right]$

$$\zeta K = \left\{\left(\frac{\det(A_1)}{\det(A)}, \frac{\det(A_2)}{\det(A)}, \ldots\right)\right\}$$ için $x_1 = \frac{\det(A_1)}{\det(A)}$ olarak bulunur. Sabit matris

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1$$
$$a_{21}x_1 + \cdots = b_2$$
$$\vdots$$
$$a_{n1}x_1 + a_{n2}x_2 + \cdots = b_n$$

$$A = \left[\begin{array}{cccc} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & & & \\ a_{n1} & a_{n2} & \cdots & \end{array}\right] \left[\begin{array}{c} x_1 \\ x_2 \\ \vdots \\ x_n \end{array}\right] = \left[\begin{array}{c} b_1 \\ b_2 \\ \vdots \\ b_n \end{array}\right]$$

$$A_1 = \left[\begin{array}{cccc} b_1 & a_{12} & a_{13} & \cdots & a_{1n} \\ b_2 & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & & & & \\ b_n & a_{n2} & a_{n3} & \cdots & a_{nn} \end{array}\right]$$

$$x+y+z=1$$
$$x-y+z=0$$
$$x+y-z=5$$

Çk cramer ile bulun

A · X = B 1. adım

$$\begin{bmatrix}1&1&1\\1&-1&1\\1&1&-1\end{bmatrix}\begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}1\\0\\5\end{bmatrix}$$

$$A_1=\begin{vmatrix}1&1&1\\0&-1&1\\5&1&-1\end{vmatrix}\xrightarrow{-5c_1+c_3\to c_3}\begin{vmatrix}1&1&1\\0&-1&1\\0&-4&-6\end{vmatrix}\xrightarrow{-3}$$

$$\Delta x=\begin{bmatrix}1&1&1\\b&-1&1\\1&1&-1\end{bmatrix}$$ 12.

$$A_2=\begin{vmatrix}1&1&1\\1&0&1\\1&5&-1\end{vmatrix}\qquad \Delta y=\begin{bmatrix}1&1&1\\1&b&1\\1&1&-1\end{bmatrix}$$

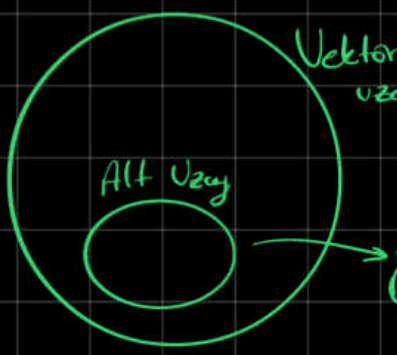
$$A_3=\begin{vmatrix}1&1&1\\1&-1&0\\1&1&5\end{vmatrix}\qquad \Delta z=\begin{bmatrix}1&1&1\\1&-1&b\\1&1&1\end{bmatrix}$$

! $$Ç_K=\left\{\left(x=\frac{|\Delta x|}{|A|}\ ,\ y=\frac{|\Delta y|}{|A|}\ ,\ z=\frac{|\Delta z|}{|A|}\right)\right\}$$ 13.

$$Ç_K=\left\{\left(\frac{\det(A_1)}{\det(A)}\ ,\ \frac{\det(A_2)}{\det(A)}\ ,\ \frac{\det(A_3)}{\det(A)}\right)\right\}$$

Vektör Uzay

Alt Uzay Vize



$R^3 \rightarrow (0,0,0)$ $(1,2,3)$ $\quad R^2 \rightarrow (0,0)$ $(1,3)$

Özel tanımlanmış alt kümedir.

Alt uzay olma şartları:

V: Vektör uzayı , W: Alt uzay

- $W_1 \in W$, $W_2 \in W$ $\rightarrow$ $W_1 + W_2 \in W$ olmalı
- $W \in W$, $c \in R$ $\rightarrow$ $c \cdot W \in W$ olmalı — ikisi de sağlanmalı

- $W = \{(x_1, x_2, x_3) \in R^3 \mid x_1 = 0\}$ kümesi R^3 ün bir alt uzayı mıdır?

bunu anla

$(0,1,3) \in W$

$(0,2,-1) \in W$

$(1,0,-1) \notin W$

1. şart: $(0, x_2, x_3) \in W$, $(0, y_2, y_3) \in W$ $\rightarrow$ $(0, x_2+y_2, x_3+y_3) \in W$ ✓ şart sağlandı

2. şart: $(0, x_2, x_2, x_3) \in W$, $c \in R$ $\rightarrow$ $c(0, x_2, x_3) = (0, cx_2, cx_3) \in W$ ✓ sağlandı

2 şart da sağlandı: W kümesi R^3 ün alt uzayıdır.

- $W = \{(x_1, x_2, x_3) \in R^3 \mid x_1 + x_2 = 2\}$ kümesi R^3 için bir alt uzayı mıdır?

$(1,1,5) \in W$
$(0,2,5) \in W$
$(1,2,5) \notin W$

- $(x_1, x_2, x_3) \in W$, $x_1 + x_2 = 2$
  $(y_1, y_2, y_3) \in W$, $y_1 + y_2 = 2$

$\hookrightarrow (x_1 + y_1, x_2 + y_2, x_3 + y_3) \notin W$

$x_1 + y_1 + x_2 + y_2 = 4 \neq 2$

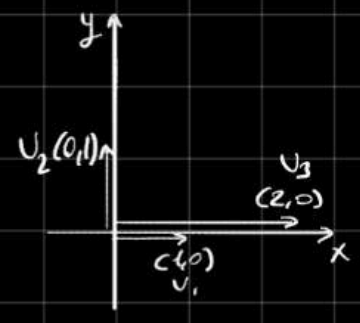
W, R^3 için alt uzayı değildir.

Lineer Bağımlılık, Lineer Bağımsızlık

$v_i \in V$, $c_i \in R$ ve $\vec{0}$ sıfır vektörü olmak üzere

$c_1v_1 + c_2v_2 + \cdots + c_nv_n = \vec{0}$ eşitliğinin tek çözümü

$c_1 = c_2 = \cdots = c_n = 0$ olması ise v_i'ler lineer bağımsızdır.



$c_1v_1 + c_2v_2 = \vec{0} = (0,0)$

$(c_1, 0) + (0, c_2) = (0,0)$

$(c_1, c_2) = (0,0)$

$c_1 = c_2 = 0$

$C.K = \{(0,0)\}$ v_1, v_2 lineer bağımsızdır.

$c_1v_1 + c_3v_3 = \vec{0}$

$(c_1, 0) + (2c_2, 0) = (0,0)$

$(c_1 + 2c_2, 0) = (0,0)$

$c_1 + 2c_2 = 0$

$C.K = \{(-2c_2, c_2) \mid c_2 \in R\}$ v_1, v_2 lineer bağımlıdır.